

Backbone Chemistry Governs the Reliability of Machine-Learned Glass Transition Temperatures: A Family-Resolved Benchmark with Novelty-Conditioned Conformal Uncertainty

Rugvedh Samala^{1,*}, Brighton Liang¹, Logan Eberhart¹

¹ [AFFILIATION: school, city, country]

* Correspondence: [CORRESPONDING EMAIL]

Abstract

Which polymer backbones can a machine-learned glass transition temperature (T_g) model be trusted on? Random partitions, reporting R^2 above 0.85, cannot answer: their test sets hold chemistry seen in training. On 7,174 repeat units in twenty backbone families plus an abstention class, histogram gradient boosting gives 28.1 K mean absolute error (random), 33.8 K (scaffold), 44.5 K (fingerprint-cluster), 48.1 K (whole-family holdout), pooled R^2 falling 0.87 to 0.75; the two best models differ by 0.38 K. The failure is chemical: withheld polyphosphazenes and polysiloxanes are overpredicted (bias -83.7 , -63.7 K), polydienes and polyolefins underpredicted ($+48.0$, $+37.8$ K); deterioration tracks structural isolation, not family size. A matched-pair design with fixed test structures attributes $+14.97$ K (95% CI $[+11.21, +19.28]$) to removing the family, worse in all twenty-one families for both models, and under 1 K of unresolved sign to removing as many unrelated structures. Uncertainty fails there: for extremely randomised trees, split conformal holds 0.904 marginal coverage (mean of splits) on random splits while covering 0.657 of the least familiar band (pooled), and loses marginal validity under shift. Family-conditioned calibration cannot repair this, a withheld family having no calibration members; similarity-conditioned calibration holds 0.877–0.907 in all four regimes (mean of splits).

1. Introduction

The glass transition temperature is the single most consulted thermal descriptor of an amorphous polymer. It bounds the service window of a thermoplastic, sets processing conditions, and governs the mechanical response of a matrix in a composite. What fixes its value is the chain itself: T_g rises with the barrier to segmental rotation about backbone bonds and with the strength of the interactions between neighbouring chains, so a rigid, strongly interacting heteroaromatic backbone vitrifies far above a flexible Si-O or P=N one. The families at the two ends of the dataset used here show that spread directly: aromatic polyimides have a median T_g of 249 °C, polysiloxanes and polyphosphazenes of -8 °C (Table 2).

Measuring T_g by differential scanning calorimetry or dynamic mechanical analysis is routine but slow, and it requires a synthesised, purified sample, so predicting it from the structure of the repeat unit has been a target of polymer theory since group-contribution methods were first formalised [1–3]. Those schemes write T_g as a composition-weighted sum of tabulated group contributions over the repeat-unit molar mass. The assumption is compositional additivity: no interaction terms, and no dependence on which linkage a group sits in. That assumption is what lets such a scheme be applied to a repeat unit its table was never fitted on, and it is also what caps its accuracy. Refitted on the training data used here, an additive scheme of this form reaches 43.8 K mean absolute error on random partitions, where a descriptor ensemble reaches 28.1 K (Section 3.9).

Data-driven surrogates have largely displaced those schemes. Curated repositories such as PoLyInfo [4] made supervised learning on experimental T_g practical, and a decade of work has explored the resulting design space: physically interpretable descriptor sets with tree ensembles and kernel methods [5,6], graph neural networks over the repeat-unit graph [7], and chemical language models pretrained on millions of hypothetical polymers [8,9]. Recent community infrastructure has consolidated the datasets and the evaluation code [10,11]. The accuracies

reported are genuinely useful: mean absolute errors in the region of 25–30 K against experimental values whose own inter-source spread can exceed 100 K.

Those numbers describe interpolation. They are obtained on random partitions, in which a test repeat unit typically has close structural analogues in training, and they therefore estimate how well a model fills in gaps within chemistry it has already seen. The motivating application is the reverse. A screening campaign enumerates candidates precisely because they are not yet in the literature, and the candidates that matter most are the ones furthest from the training distribution. The mismatch between the validation regime and the deployment regime is well known in cheminformatics, where scaffold and time-based splits were introduced for exactly this reason [12,13], and it has begun to be examined for polymers [14,15].

Three questions remain open, and this work addresses all three.

Which backbone chemistries does a model fail on, and in which direction? An aggregate error over a heterogeneous collection is not actionable for a chemist working on one class of polymer. Errors resolved by family, and reported with their sign, say what a pooled MAE cannot: whether a backbone is placed systematically too high or too low, and whether that displacement follows the chemistry or merely follows the distance of the family's T_g from the rest of the record. The two readings have to be separated before the error structure can be called chemical.

Is the family-holdout penalty caused by the missing chemistry? Withholding an entire polymer family and observing an error increase is the standard diagnostic, but it changes two things at once. The test population changes — families differ in how intrinsically hard they are and in how wide their T_g distribution is — and the training set shrinks. A comparison against a random-split baseline therefore measures the sum of a population effect, a data-volume effect and the effect of interest. The closest prior analysis of this dataset states the confound explicitly and leaves it unresolved [16]. Resolving it requires holding the test set fixed and matching the training pools on size.

****Can a T_g prediction for a backbone the model has not seen be trusted?*** A point prediction without a trustworthy interval is not actionable for screening, and split conformal prediction [17–19] is an attractive answer because it wraps any regressor in intervals with a finite-sample coverage guarantee under exchangeability. Two difficulties arise here. Exchangeability is precisely what a family holdout violates. And the guarantee is *marginal*: a predictor can achieve 90% coverage overall while systematically under-covering a subpopulation, and in this problem that subpopulation is the unfamiliar chemistry the model was deployed to explore. Ensemble uncertainty paired with an explicit applicability-domain check has been applied to polymer property prediction [20], but without a family-resolved conditional-coverage test. A recent analysis on a 410-sample simulation-derived set reports this pattern and characterises conformal intervals as conservative triage indicators rather than fine-grained screening tools [15]; whether the failure is repairable, and whether it persists at the scale and heterogeneity of the experimental record, has not been tested.

1.1. Contributions

1. **An independently re-derived, chemically resolved dataset.** We re-curate 7,208 experimental repeat-unit records to 7,174 canonical structures, reproducing a prior public analysis of this collection [16] step for step, and assign each structure to one of twenty backbone families or to an explicit abstention class using a *backbone-aware* substructure classifier. Classifying on the chain path rather than on whole-molecule matches keeps poly(alkyl acrylate)s out of the polyester class, which matters because every family-holdout conclusion inherits the labels.

2. ****A family-resolved account of where prediction fails, and in which direction.*** Holdout deterioration spans a factor of nearly three across the twenty-one groups, and the signed error is systematic: withheld polyphosphazenes and polysiloxanes are

overpredicted, by 83.7 K and 63.7 K on average, and withheld polydienes and polyolefins underpredicted, by 48.0 K and 37.8 K. Regression toward the training mean accounts for part of the first pair and none of the second, so the chemical reading rests on the underpredicted hydrocarbon families rather than on the inorganic ones (Section 3.3). Deterioration tracks how structurally isolated a family is (Pearson r between -0.62 and -0.69 across four models) while a family-size effect is not resolved at twenty-one families.

3. A matched-pair design that isolates the family effect. Three training pools — one retaining the family, one with it removed, and one size-matched control from which an equal number of unrelated structures was removed — predict an identical held-out test set, separating the cost of missing chemistry from the cost of missing data.

4. ****A test of whether a T_g prediction for an unseen backbone can be trusted****, by conditional-coverage evaluation of conformal uncertainty, with a novelty-conditioned Mondrian taxonomy that substantially narrows the subgroup coverage gap where a family-conditioned taxonomy structurally cannot.

5. An applicability-domain diagnostic in units a practitioner can act on.

Using conformal prediction itself as the applicability-domain statement, in the tradition established by Norinder et al. [21], coverage, interval width and absolute error are resolved by nearest-neighbour Tanimoto similarity, and the similarity itself is reported alongside each prediction so that a reader can see which regime a prediction sits in. We do not offer a single similarity below which an interval should be disbelieved, because under family holdout the error is not monotone in similarity: for extremely randomised trees it falls from 55.5 K below Tanimoto 0.4 to 30.3 K in the 0.6–0.7 band and then rises again to 36.7 K above 0.8 (Section 3.5), so no cut-point separates trustworthy from untrustworthy predictions.

6. ****A Van Krevelen group-contribution reference point and a taxonomy sensitivity check****, so that the machine-learned result is placed against the additive schemes it proposes to replace and against a perturbation of its own labels.

7. A reproducible open implementation, with leak-free in-fold preprocessing and a test suite pinning both the chemistry of the family assignment and the finite-sample behaviour of the conformal predictors.

2. Materials and Methods

2.1. Dataset and curation

We use the experimental homopolymer T_g collection redistributed with the POINT² polymer informatics benchmark [11], which derives from the PoLyInfo record [4]. Each entry pairs a repeat unit written as a PSMILES string — SMILES with two `\*` atoms marking the polymerisation attachment points — with a reported T_g .

The raw export is record-level: one repeat unit may appear several times carrying values from different primary sources. Modelling on records would allow identical inputs to fall on both sides of a partition and would weight frequently reported polymers more heavily, so all analysis is performed on a structure-level table. Repeat units were canonicalised with RDKit [22] and grouped; where a canonical structure carried several reported values, the median was taken, and the number of contributing records and the full reported range were retained as provenance.

2.2. Backbone-aware family assignment

Polymer family labels were derived rather than inherited, so that the taxonomy is reproducible and its chemistry is inspectable. For each repeat unit the *backbone* is taken as the union of shortest paths between attachment points, extended to include any ring the path enters.

Characteristic linkages are then matched against SMARTS patterns and accepted only when their core atoms lie on that backbone.

Because a repeat unit may be written cut at its own characteristic linkage — nylon-6 as ``*NCCCCC(=O)*`` divides its amide between the two ends — the classifier operates on a head-to-tail dimer rather than on the single unit. Joining two copies restores the linkage the cut destroyed while keeping the junction a bond that exists in the real chain.

Backbone restriction is what separates a poly(alkyl acrylate) from a polyester: both contain an ester group, but in the acrylate it decorates an all-carbon chain. Patterns are tested in order of specificity, because the groups are nested — every cyclic imide contains two amide-like motifs, and a urethane contains both ester-like and amide-like fragments. Repeat units whose backbone carries no heteroatom linkage are classified by the groups pendant to the chain (aromatic-dominated, acrylic, styrenic, halogenated, other heteroatom-substituted vinyl, unsaturated, or saturated hydrocarbon). A unit whose backbone does carry a heteroatom but matches no linkage pattern is assigned to an explicit abstention class, **Other backbone**, rather than forced into the nearest hydrocarbon category. The twenty-five chemistry cases in Table S1 are pinned by unit tests.

2.3. Representations

Two representations were computed. The full RDKit descriptor block (217 columns) provides physically interpretable features; the topological ``Ipc`` index was replaced by its base-10 logarithm because it overflows for larger repeat units. Binary Morgan fingerprints (radius 2, 2,048 bits) [23] provide the similarity space in which structural novelty is measured. Attachment points were retained in both, since they identify where the chain continues and the environments around them encode exactly the linkage motifs that set T_g .

Descriptor column filtering — removal of non-finite, constant and exactly duplicated columns — is implemented as the first step of a scikit-learn pipeline [24] and is therefore **fitted on the training fold only**. Performing this filtering once over the full table, as is common, lets test structures influence the feature definition; the closest prior analysis of this dataset notes this as an unquantified limitation of its own results [16].

2.4. Partitioning regimes

Four regimes were used, in increasing order of enforced novelty: **random** partitions (repeated, 80:20); **scaffold** partitions, in which generic Bemis-Murcko scaffolds [25] are assigned whole; **cluster** partitions, using average-linkage agglomerative clusters in Jaccard space over Morgan bits; and **family** holdouts, in which every member of one backbone family is withheld. All non-family regimes were repeated ten times with different seeds, because single-split rankings among tree ensembles on this dataset fall inside run-to-run noise. The family regime contributes twenty-one splits, one per family.

2.5. Matched-pair design

A family is eligible if it holds at least forty structures, which all twenty-one groups in this dataset do. For each eligible family F , a test set $T \subset F$ is drawn (30% of F) and three training pools are constructed: **informed** (everything but T , so the remainder of F is available), **naive** (everything but F entirely), and **control** (**informed** minus a randomly chosen block of structures from other families, equal in number to $F \setminus T$). All three predict the same T . The draw of T and of the control block is repeated three times per family with different seeds, giving sixty-three matched triples per model.

The contrast **naive** – **control** is the quantity of interest. Both pools are identical in size and both exclude T ; they differ only in *which* structures were removed. The contrast **control** – **informed** measures the pure data-volume effect over the same test set.

Two properties of the design bound what it can establish. First, the removed block is small: it is 70% of one family, a median of 1.85% of the informed pool across the twenty-one families, ranging from 0.43% for polyureas to 17.94% for polyimides, with sixteen of twenty-one families under 5%. Second, the two arms remove the same *quantity* but not the same *kind* of structure: the naive arm removes a coherent chemical neighbourhood, the control arm a diffuse random sample spread over the remaining families. The contrast therefore measures the effect of removal *locality* at fixed volume. That chemistry is what locality means here is an interpretation, argued in Section 4.2, not a separate measurement.

2.6. Models

Five regressors were evaluated on the descriptor representation: a training-median baseline, support vector regression, random forest, histogram gradient boosting and extremely randomised trees. Every model is wrapped in the same pipeline so that all data-dependent preprocessing is fitted in-fold.

A sixth model, a group-contribution baseline, was run through the same four regimes. It implements the classical Van Krevelen additive form $T_g = Y_g/M$ with $Y_g = \sum n_i Y_i$, where M is the repeat-unit molar mass, n_i the count of structural group i and Y_i its molar glass-transition contribution [1-3]. Groups are RDKit fragment counts supplemented by element, ring, rotatable-bond, hydrogen-bonding and backbone-length counts, which carry the chain skeleton that fragment counts alone would miss. The contributions Y_i are fitted by ridge regression on the training fold rather than taken from a published table, so the additive scheme receives the same data the ensembles receive and any difference between them reflects the functional form rather than the calibration set. Fitting is performed in $Y_g = T_g \cdot M$ space and in absolute temperature, since an additive law expressed in Celsius would make the contributions depend on an arbitrary zero.

2.7. Conformal prediction and coverage diagnostics

The screening question this machinery is built to answer is a narrow one: given a predicted T_g for a repeat unit whose backbone chemistry the model has not seen, how wide an interval has to be quoted before the number can be acted on? The construction below is standard; what it is asked to do here is chemical.

Each training pool is divided three ways: a **fit** partition (60%) trains the point regressor, a **difficulty** partition (20%) supplies out-of-sample residuals, and a **calibration** partition (20%) supplies the conformal quantile. Fitting the difficulty model on the point model's own training residuals would teach it that the model is everywhere accurate, so that partition must be held out. All conformal variants share one point predictor and one calibration set, so differences between them reflect only how residuals are converted into intervals.

The three-way division costs accuracy. On a random 80:20 partition the point regressor is fitted on about 48% of the dataset, three-fifths of what an 80%-trained literature model sees, so the random-split errors reported below are mildly pessimistic relative to published figures obtained without a calibration carve-out. Data-reusing constructions such as the jackknife+ avoid the carve-out at the cost of refitting [26]; the split construction is kept here so that all four interval methods share one fitted model and one calibration set and differ in nothing else.

Given calibration scores $s_1 \dots s_n$ and a miscoverage level α , the conformal quantile is the $[(n+1)(1-\alpha)]$ -th order statistic; when that index exceeds n the honest interval is unbounded. We report $\alpha = 0.1$ throughout. Three predictors are compared:

- **SCP** — one global quantile of absolute residuals.
- **Normalised SCP** — residuals scaled by a fitted difficulty estimate, so intervals widen where the model expects to struggle rather than everywhere [27].
- **Mondrian SCP** — a separate quantile per category [28,29],

under two taxonomies: *polymer family*, and *nearest-training-similarity band*.

The asymmetry between the two taxonomies is the point. A family-conditioned taxonomy is undefined for a family with no calibration members, which is exactly the family-holdout case; a novelty-conditioned taxonomy is defined for any repeat unit, seen family or not. A third route to validity under shift is to reweight the calibration scores by the likelihood ratio between the test and training covariate distributions [30]; Section 4.3 says why that route is not taken here.

A fourth, an **oracle inflation control**, is used only to test whether the novelty-conditioned predictor's advantage reduces to its intervals being wider. It multiplies every split conformal interval about its centre by a single constant k . Two choices of k are reported. The **coverage-matched** constant reproduces the novelty-conditioned predictor's marginal coverage on the same test set, and is the primary control, because it fixes the coverage budget and leaves only its allocation across the similarity range to differ. The **width-matched** constant equalises the mean interval width instead; it answers a different question, and it hands the inflated predictor more marginal coverage than the predictor it is compared against. Both are oracles and neither is a deployable method, because k is read off the test set it is then compared against; their role is to bound what *any* global inflation of split conformal could achieve.

Alongside marginal coverage and mean interval width we report **conditional** coverage within each similarity band and each family, the **maximum coverage gap** (the largest shortfall below nominal across subgroups with at least ten members), and the coverage-width criterion. Ensemble spread is additionally assessed by sparsification error.

Coverage can be aggregated in four ways that differ here by several points, so every coverage figure below is labelled with the one used.

- **Pooled** marginal coverage is the fraction of all held-out predictions in a regime that fall inside their interval, taking every split together. It is the quantity a single large test set would give.
- **Mean-of-splits** marginal coverage is the per-split coverage averaged over splits, so a 44-structure family weighs as much as a 1,707-structure one. Under family holdout split conformal gives 0.701 this way against 0.743 pooled.
- **Band-resolved** coverage is pooled within a similarity band across splits, since a single split may contribute few or no structures to a band.
- **Split-averaged worst-band** coverage takes the lowest band coverage within each split, over bands holding at least ten structures, and averages those minima. Taking the minimum before the average makes it lower than the band-resolved coverage of the least-similar band. The size restriction is material: dropping it lowers the split conformal figure of Table 13 from 0.564 to 0.539, because a band of two or three structures can read 0.0 on binomial noise alone.

The convention adopted is mean-of-splits for marginal coverage and band-resolved for coverage within a similarity band. Two places depart from it and say so: the split-averaged worst band is used in Figure 4b and in the calibration-construction comparison of Section 3.8, and the oracle controls of Section 3.4 are quoted pooled throughout, because they compare two predictors structure by structure on the same held-out set and a split-level average would discard that pairing.

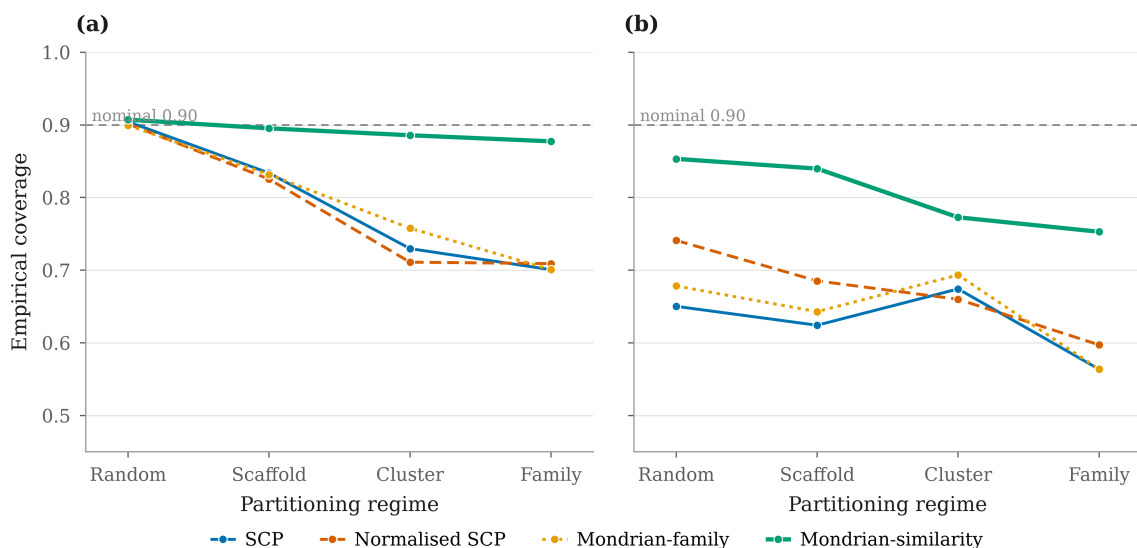


Figure 4. Coverage of nominally 90% conformal intervals, extremely randomised trees. (a) Marginal coverage, as a mean over the splits of each regime. (b) The split-averaged worst band: within each split the weakest similarity band holding at least ten structures is taken and those minima are averaged. Line style encodes method alongside colour, so the exact coincidence of split conformal and the family-conditioned quantile under family holdout remains visible.

The coverage-width criterion (CWC) is the mean interval width multiplied by $\exp(\eta(1 - \alpha - \text{coverage}))$ when coverage falls below nominal and left unpenalised otherwise, with $\eta = 30$. Because the penalty is exponential in the shortfall, CWC differences between methods are dominated by that exponent and not by width: the criterion is close to a recoding of the coverage gap already reported, and it is quoted only for comparability with the conformal literature.

2.8. Statistical treatment

Confidence intervals on the matched-pair effects are percentile bootstrap intervals over the twenty-one family-level means, each of which averages the three repeats. Those twenty-one values are neither independent nor a sample. The naive and control pools of a given family share at least 78% of their training data by construction — the minimum is polyimides, the largest family, and the figure exceeds 95% in sixteen of the twenty-one families; the families are chemically nested, so that polycarbonates sit inside the same carbonyl chemistry as polyesters; and the twenty-one are the entire population of families in this dataset rather than a draw from a larger one. The intervals should therefore be read as descriptive dispersion across families, not as inference about a population of families.

The Wilcoxon signed-rank test on twenty-one paired values has a smallest attainable two-sided p of $2^{-20} = 9.5 \times 10^{-7}$, reached whenever all twenty-one signs agree. Where that value appears below it is the floor of the test, not a measured quantity, and the informative statement is the sign count.

p -values are reported unadjusted for multiplicity. The one place where this matters is the driver analysis of Section 3.3, which runs sixteen correlations (four candidate drivers $\times$ four models). Under a Bonferroni correction at $\alpha = 0.05/16 = 0.0031$, the nearest-training-similarity correlation survives for all four models and no other driver survives for any.

2.9. Reproducibility

All code, the curated structure-level table and every result file are available at <https://github.com/Rugvedhs/spectrasense>, in the directory ``polymer-tg-benchmark`. Seeds are fixed, and every figure and every result CSV is written by a script from the stage outputs, the oracle controls of Section 3.4 and the shrinkage regression of Section 3.3 included: eleven of the fourteen table sources by `scripts/analyze.py`, the curation and composition tables by `scripts/prepare_features.py` and the calibration study by `scripts/run_calibration_study.py`. A cold

run reproduces every number in this paper. The fourteen tables typeset in the body are transcriptions of the CSV named in each caption. `scripts/check_manuscript_tables.py` re-parses those tables out of the manuscript source and asserts every cell against its named CSV at the rounding quoted; it runs as part of the test suite.

3. Results

3.1. The public record survives re-derivation

Independent re-curation of the 7,208 raw records reproduces a prior public analysis of this collection [16] step for step, and Table 1 sets the two side by side so that the correspondence can be read rather than asserted: 7,174 unique raw SMILES, no RDKit parse failures, 7,174 canonical structures, 31 canonical groups containing more than one record, 34 records absorbed by aggregation, 28 of those groups carrying disagreeing T_g values, and a maximum within-structure range of 115 K. Attachment-point counts likewise agree once the level is matched: that analysis counts 7,204 records with two attachment points, three with three and one with four, and we count 7,170 structures with two, three with three and one with four, the difference being the 34 duplicate records aggregated away. The dataset is therefore a stable object to build on, and the numbers below are not sensitive to curation choices.

Table 1. Curation audit of the POINT² experimental T_g export [11]: reported by [16] | this work. Counts describe the whole collection, not a model run; the within-structure range is the largest spread between independent reports of one canonical repeat unit. The attachment-point rows are record-level in the prior analysis [16] and structure-level here, which is the whole of their disagreement. The values in the prior-analysis column are transcribed from that work and are not an output of this pipeline; our column is `results/table_curation.csv` and `results/table_attachment_points.csv`.

Step	Reported by [16]	This work
Raw records	7,208	7,208
Unique raw SMILES	7,174	7,174
RDKit parse failures	0	0
Canonical structures retained	7,174	7,174
Structures aggregated from >1 record	31	31
Records absorbed by aggregation	34	34
Aggregated groups with disagreeing T_g	28	28
Max within-structure T_g range (K)	115	115
Units with 2 / 3 / 4 attachment points	7,204 / 3 / 1 (records)	7,170 / 3 / 1 (structures)

That 115 K figure deserves emphasis before any model error is quoted. It is the spread between independent literature reports of the *same* repeat unit, and it places a floor under what structure-based prediction can achieve: a repeat-unit representation cannot resolve differences in molecular weight, tacticity, crosslinking, thermal history or measurement protocol, and those differences are folded into the target.

The derived taxonomy places all twenty-five reference polymers of Table S1 in the family a polymer chemist would assign, and yields the family composition of Table 2. The resulting family T_g ordering (Figure 1) is likewise the expected one, which is a further check that the labels are chemically meaningful rather than merely self-consistent: among the named families, polyphosphazenes (median -8 °C) and polysiloxanes (-8 °C) sit lowest, reflecting the low rotational barriers of P=N and Si-O backbones, and aromatic polyimides sit highest (249 °C), reflecting rigid, strongly interacting heteroaromatic chains. The only group below them is the abstention class Other backbone (median -19.6 °C, 58 structures), of which 57 contain silicon; its position is consistent with the silane and carbosilane chemistry it holds.

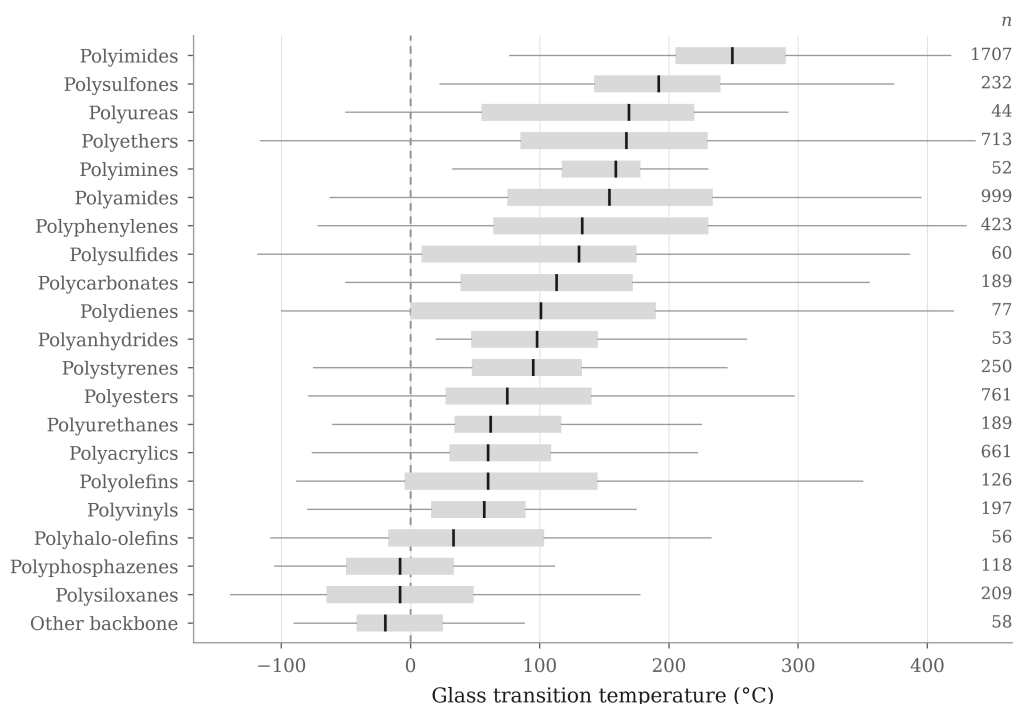


Figure 1. Reported glass transition temperature by repeat-unit family, ordered by median; boxes span the interquartile range, whiskers the 5th to 95th percentiles, and family sizes are listed at the right. The ordering is the one polymer chemistry predicts, which is evidence that the derived taxonomy is meaningful rather than merely self-consistent.

Table 2. Composition of the twenty backbone families and the *Other backbone* abstention class, over the 7,174 structure-level records; T_g in °C as reported, ordered by family size. Means and standard deviations are in results/table_families.csv.

Family	<i>n</i>	Median T_g (°C)	Min (°C)	Max (°C)
Polyimides	1707	249.0	-9.0	490.0
Polyamides	999	154.0	-62.0	395.0
Polyesters	761	75.0	-78.6	400.0
Polyethers	713	167.0	-116.0	437.0
Polyacrylics	661	60.0	-100.0	330.0
Polyphenylenes	423	133.0	-71.4	495.0
Polystyrenes	250	95.0	-75.0	330.0
Polysulfones	232	192.0	23.0	422.0
Polysiloxanes	209	-8.0	-139.0	177.5
Polyvinyls	197	57.0	-79.5	174.6
Polyurethanes	189	62.0	-60.0	225.0
Polycarbonates	189	113.0	-50.0	355.0
Polyolefins	126	60.0	-88.0	350.0
Polyphosphazenes	118	-8.0	-105.0	291.0
Polydienes	77	101.0	-99.6	420.0
Polysulfides	60	130.5	-118.0	386.0
Other backbone	58	-19.6	-90.0	152.0
Polyhalo-olefins	56	33.2	-108.0	232.5
Polyanhydrides	53	98.0	20.0	260.0
Polyimines	52	159.0	10.0	327.0
Polyureas	44	169.0	-50.0	292.0

3.2. Accuracy is governed by the split, not by the model

On random partitions the best model, histogram gradient boosting, reaches 28.1 ± 1.0 K MAE ($R^2 = 0.87$), consistent with published values for descriptor-based ensembles on this dataset once the calibration carve-out of Section 2.7 is allowed for. That ranking is fragile. Across ten repeated splits, histogram gradient boosting beats extremely randomised trees by 0.38 K, consistently in sign (9 of 10 splits, Wilcoxon $p = 0.049$) but negligible in magnitude; a paired t -test does not resolve it ($p = 0.069$) because one repeat reverses. The magnitude, not the significance, is what matters: 0.38 K against a 20.0 K protocol effect. Random forests and support vector regression are worse by 2.17 K and 2.74 K ($p < 0.0001$ each). Reporting a single split, as is common, would allow either of the leading models to be declared the winner.

Against that sensitivity, the choice of partition moves the error by far more (Figure 2, Table 3). Holding the model at histogram gradient boosting, MAE rises from 28.1 K on random splits to 33.8 K on scaffold splits, 44.5 K on fingerprint-cluster splits and 48.1 K under whole-family holdout. The random-to-family contrast is therefore 20.0 K: about fifty times the 0.38 K that separates the two best models, and about seven times the 2.74 K that spans all four non-trivial regressors. Only the training-median baseline, at +65.7 K, is moved more by the choice of model than by the choice of protocol. Whatever a leaderboard on this dataset is ranking, it is not the property that changes most.

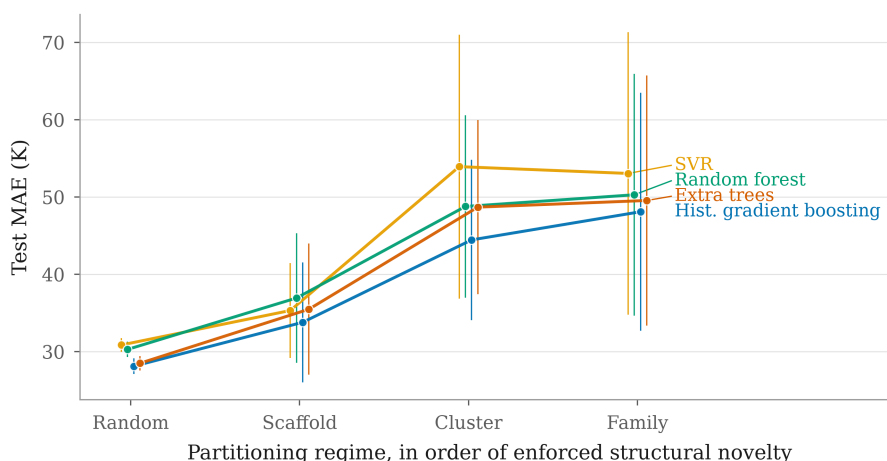


Figure 2. Test MAE in K by model and split regime; error bars are the standard deviation across splits. For random, scaffold and cluster that is spread across ten repeats of the same protocol; for family holdout each family is withheld exactly once, so the bar is dispersion between families and not run-to-run noise. The spread between regimes dwarfs the spread between models.

Table 3. Test mean absolute error in K by model and split regime. Each cell is the mean over splits, with the standard deviation across splits after $\pm$. Random, scaffold and cluster are ten repeated splits; family holdout contributes twenty-one splits, one per family, so its spread is dispersion between families rather than run-to-run noise. The last row is the mean nearest-training Tanimoto similarity of the held-out structures. RMSE and R^2 per cell are in results/table_point_by_regime.csv.

Model	Random (10)	Scaffold (10)	Cluster (10)	Family (21)
Histogram gradient boosting	28.13 ± 1.01	33.81 ± 7.76	44.47 ± 10.37	48.11 ± 15.39
Extremely randomised trees	28.51 ± 0.93	35.50 ± 8.48	48.71 ± 11.27	49.56 ± 16.19
Random forest	30.30 ± 1.00	36.95 ± 8.37	48.79 ± 11.81	50.30 ± 15.66
Support vector regression	30.88 ± 0.90	35.34 ± 6.15	53.94 ± 17.06	53.06 ± 18.28
Training-median baseline	93.79 ± 1.65	95.71 ± 13.67	121.61 ± 24.23	95.46 ± 28.61
Mean nearest-training similarity	0.762	0.671	0.403	0.454

Mean nearest-training Tanimoto similarity falls alongside the error, from 0.76 on random splits to 0.67, 0.40 and 0.45. That ordering is not strictly monotone: family holdout leaves test structures slightly *less* isolated on average (0.45) than cluster holdout (0.40), because a withheld family may still have near analogues in a chemically adjacent one, yet family holdout produces the higher error. Structural distance alone therefore does not determine difficulty; *which* chemistry is missing also matters, a point Section 3.3 takes up directly.

R^2 must be quoted with its aggregation stated, because the two natural choices disagree by more than any effect in this paper (Table 4). Pooling residuals over all held-out structures in a regime, which is the quantity comparable to a random-split R^2 , gives 0.87 on random splits, 0.81 on scaffold, 0.68 on cluster and 0.75 under family holdout. Averaging R^2 over the twenty-one individual family holdouts instead gives 0.26, because each family is scored against its own T_g variance and a family with a narrow T_g spread contributes a large negative value at a modest absolute error. That 0.26 is a real number but it is not commensurable with 0.87, and the fall from 0.87 to 0.26 that a naive comparison suggests is an aggregation artefact. The comparable fall is 0.87 to 0.75.

Table 4. The two aggregations of R^2 , for histogram gradient boosting. Pooled R^2 is computed once over all held-out predictions in a regime; the mean-of-splits value averages the per-split R^2 , each scored against its own split's T_g variance. The two are not interchangeable, and the family-holdout row is where they diverge. All five models are in results/table_pooled_r2.csv.

Regime	Predictions pooled	Pooled R^2	Mean of per-split R^2
Random (10 splits)	14,350	0.866	0.865
Scaffold (10 splits)	14,863	0.808	0.792
Cluster (10 splits)	24,985	0.679	0.538
Family (21 splits)	7,174	0.746	0.256

The dispersion is as informative as the mean. Random-split MAE varies by ± 1.0 K across ten repeats, and scaffold and cluster splits by ± 7.8 K and ± 10.4 K; these are repeat-to-repeat quantities. The ± 15.4 K quoted for family holdout is not: each family is held out exactly once, so it is the standard deviation *between families*, and it describes how unevenly the penalty is distributed over chemistry rather than how reproducible any one holdout is. Either way, a point estimate from one chemistry-aware split should not be trusted.

3.3. Transfer failure is a property of backbone chemistry

Family-holdout deterioration, defined as the ratio of family-holdout MAE to random-split MAE for the same model, spans a factor of nearly three across the twenty-one families (Table 5; histogram gradient boosting throughout this section). At the difficult end sit polyphosphazenes ($3.02\times$, 84.8 K MAE), polydienes ($2.83\times$, 79.6 K) and polysiloxanes ($2.48\times$, 69.7 K). At the easy end, polyvinyls deteriorate by $1.07\times$, polyethers by $1.19\times$ and polycarbonates by $1.20\times$; no family is predicted better when withheld than the average random-split structure, though polyvinyls come within 8%.

Table 5. Family-holdout error for histogram gradient boosting: the three most and three least deteriorated of the twenty-one families. Each family is withheld exactly once, so n is the number of held-out structures and there is no repeat-to-repeat spread. Deterioration is the family's holdout MAE divided by the same model's mean random-split MAE of 28.13 K. All twenty-one families are listed in Appendix A and in results/table_family_holdout.csv.

Family	n	MAE (K)	Deterioration	Mean NN similarity
Polyphosphazenes	118	84.8	$3.02\times$	0.304
Polydienes	77	79.6	$2.83\times$	0.367
Polysiloxanes	209	69.7	$2.48\times$	0.388
...				

Family	<i>n</i>	MAE (K)	Deterioration	Mean NN similarity
Polycarbonates	189	33.7	1.20×	0.558
Polyethers	713	33.4	1.19×	0.553
Polyvinyls	197	30.2	1.07×	0.430

The signed error carries the mechanism, but it has to be separated from a purely statistical effect first. Defining bias as $\text{mean}(T_g \text{ observed} - T_g \text{ predicted})$, so that a negative value is overprediction, the withheld inorganic backbones are displaced by wide margins: polyphosphazenes by -83.7 K, polysiloxanes by -63.7 K and the Other backbone silane class by -45.3 K, against MAEs of 84.8, 69.7 and 52.5 K. Across the twenty-one families the median ratio of $|\text{bias}|$ to MAE is 0.54, so about half of a typical family-holdout error is systematic displacement rather than scatter, and its magnitude tracks structural isolation, the absolute bias correlating with mean nearest-training similarity at $r = -0.53$ ($p = 0.014$, unadjusted).

A model fitted almost entirely on other families will pull any withheld family toward the bulk of the training distribution, so a family whose T_g sits far from that bulk is displaced toward it for reasons that have nothing to do with its backbone. Regressing bias on the family's T_g offset from the rest of the dataset measures how much of the signal that accounts for (Table 6). The two are positively correlated, at $r = +0.425$ ($p = 0.055$, twenty-one families): marginal, but large enough to matter here, because it is the three families just quoted that the line fits. Polyphosphazenes sit 145 K below the rest of the dataset, polysiloxanes 148 K below and the silane class 155 K below, and shrinkage alone accounts for -23.4 K, -24.0 K and -25.3 K of their bias, which is 28%, 38% and 56% of what is observed. They remain the most displaced families in absolute terms, but they cannot carry the chemical argument by themselves.

The families that can carry it are the ones moving against the shrinkage line. Polydienes (bias $+48.0$ K, residual $+49.9$ K), polyolefins ($+37.8$ K, $+47.7$ K), polyhalo-olefins ($+28.4$ K, $+43.5$ K) and polystyrenes ($+19.9$ K, $+23.6$ K) are all *under*predicted even though every one of them sits below the dataset median, where regression toward the training mean would push the prediction up rather than down. A model deprived of its flexible unsaturated and saturated hydrocarbon chains places them too low, having learned its temperature scale from stiffer condensation backbones, and no shrinkage toward the training mean produces that sign. The chemical reading of the signed error therefore rests on these four families rather than on the inorganic ones, and this line of evidence is partial: for the families with the most extreme T_g the two explanations are not separated by the data available here.

Table 6. Family-holdout bias against regression toward the training mean, histogram gradient boosting, one row per withheld family. Offset is the family's median T_g minus the median of the rest of the dataset; bias is $\text{mean}(T_g \text{ observed} - T_g \text{ predicted})$ over the withheld family; both in K. Expected bias is the least-squares fit of bias on offset across all twenty-one families (Pearson $r = +0.425$, $p = 0.055$), and residual is bias minus that fit. Shown are the four families with the largest positive residuals, the four with the largest negative, and the abstention class. All twenty-one are in results/table_bias_shrinkage.csv.

Family	Offset (K)	Bias (K)	Expected from offset (K)	Residual (K)
Polydienes	-34.0	+48.0	-1.9	+49.9
Polyolefins	-75.0	+37.8	-9.9	+47.7
Polyhalo-olefins	-101.8	+28.4	-15.1	+43.5
Polystyrenes	-43.0	+19.9	-3.7	+23.6
Other backbone	-154.6	-45.3	-25.3	-20.0
Polyimides	+150.5	+3.9	+33.8	-29.9
Polysiloxanes	-148.0	-63.7	-24.0	-39.6
Polyureas	+35.0	-43.1	+11.4	-54.5
Polyphosphazenes	-145.0	-83.7	-23.4	-60.3

Polycarbonates illustrate the benign case from the same angle: they are surrounded in structure space by polyesters and polyethers, which remain in training and carry nearly the same carbonyl and ether linkages, so their withdrawal removes little the model cannot recover, and their bias is a modest +15.8 K.

Four candidate explanations were tested against deterioration across the twenty-one families, and only structural isolation survives. Mean nearest-training similarity correlates strongly and negatively for every model (Pearson r between -0.62 and -0.69 , $p \leq 0.0027$, significant for all four models after Bonferroni correction over the sixteen tests). The shift between the family's median T_g and the rest of the dataset gives Pearson r between 0.32 and 0.39 ($p \geq 0.079$), and the ratio of the family's T_g interquartile range to the dataset's gives r between 0.15 and 0.32 ($p \geq 0.16$); neither is resolved here. Family size is likewise not resolved, at r between -0.05 and -0.35 ($p \geq 0.115$; for extremely randomised trees, $r = -0.27$, 95% CI $[-0.63, +0.18]$). That interval is wide enough to contain a moderate effect in either direction, so the honest statement is that a size effect is undetectable at twenty-one families, not that it is absent.

One feature of the design protects the headline contrast. Mean nearest-training similarity and family size are nearly orthogonal across the twenty-one families ($r = 0.10$, $p = 0.67$), so the similarity result is not a relabelled size result and the two candidate drivers are not competing for the same variance.

3.4. Conformal intervals lose validity exactly where they are needed

All conformal results below use extremely randomised trees as the point predictor, not the histogram gradient boosting of Section 3.2. Of the two leading models it is the only one that exposes a per-tree ensemble spread, which the sparsification diagnostic of Section 2.7 needs, and the accuracy it gives up is 0.38 K of random-split MAE, consistent in sign across nine of the ten repeats but small against every effect this paper measures (Section 3.2). The paper's two headline results therefore come from different models: the 28.1 K to 48.1 K accuracy gap from histogram gradient boosting, every coverage figure from extremely randomised trees.

Split conformal behaves as advertised on random partitions: 0.904 empirical coverage against a nominal 0.90, as a mean over the ten splits. Under shift the guarantee degrades with structural novelty — 0.834 on scaffold splits, 0.730 on cluster splits and 0.701 under family holdout, all mean-of-splits (Figure 4a; Table 7). This is not a defect of the method but of its premise: conformal validity requires calibration and test data to be exchangeable, and a chemistry-aware holdout is constructed precisely to break that.

Marginal coverage also conceals a failure that is present even when it holds. On random splits, where overall coverage is a healthy 0.904, coverage among the structures whose nearest training analogue lies below Tanimoto 0.4 is 0.657 pooled over the ten splits (Figure 5); the split-averaged worst band, which takes each split's weakest band before averaging, is 0.650 (Figure 4b). A nominally 90% interval covers two-thirds of the repeat units that a screening campaign would actually be evaluating. A model can pass every validity check in current practice and still be wrong about a third of the unfamiliar polymers it is deployed on.

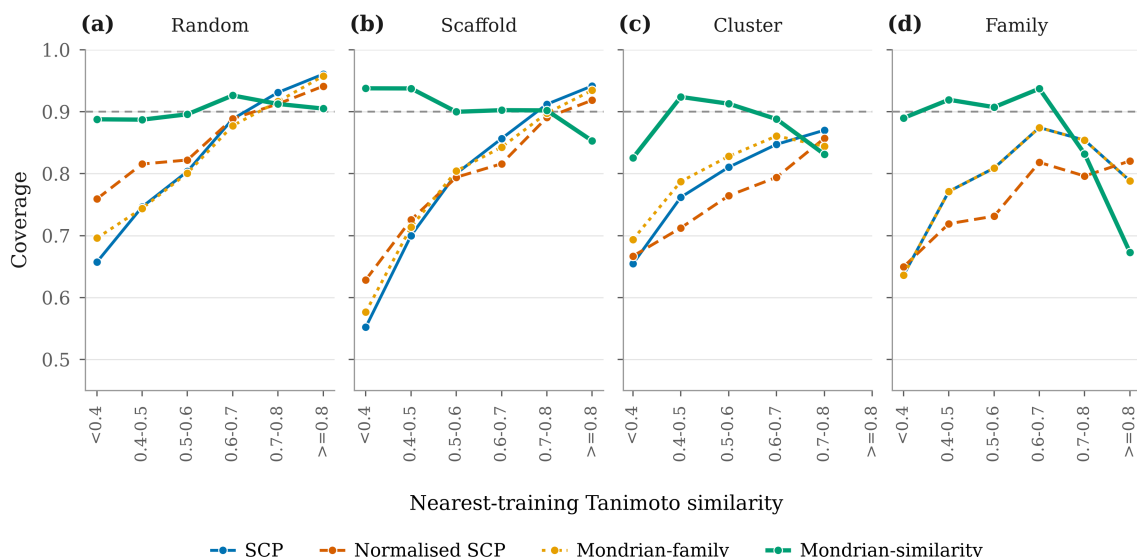


Figure 5. Conditional coverage against nearest-training Tanimoto similarity, one panel per split regime, pooled within each band across the splits of that regime. Split conformal degrades monotonically with structural novelty; the novelty-conditioned quantile stays near nominal.

Neither obvious remedy works. The normalised predictor, which scales residuals by a fitted difficulty estimate, does lift the least-similar band: from 0.657 to 0.759 on random splits and from 0.552 to 0.629 on scaffold splits, both band-resolved. Its marginal coverage still falls away under shift, to 0.825 on scaffold, 0.711 on cluster and 0.709 under family holdout, mean-of-splits, and under cluster holdout that is the worst marginal coverage of the four methods. Conditioning the quantile on polymer family fails for a structural reason rather than an empirical one: under family holdout the withheld family has no calibration members by construction, so the predictor falls back to the global quantile for 100% of test structures and reproduces split conformal exactly (0.701 mean-of-splits, identical interval widths).

Conditioning on nearest-training similarity is the variant that survives, because that coordinate is defined for every repeat unit whether or not its family was seen. It holds 0.907, 0.895, 0.886 and 0.877 mean-of-splits across the four regimes, never requiring a fallback under family holdout. On random splits it achieves this while being *narrower* than split conformal (129.1 K against 132.3 K), and it widens only where the chemistry demands it, to 155 K under scaffold, 232 K under cluster and 208 K under family holdout (Table 7). The coverage-width criterion registers the same thing, at 411 under family holdout against 48,020 for split conformal, but that ratio is almost entirely the exponential penalty of Section 2.7 acting on a 0.20 coverage shortfall rather than a statement about width, and it should be read as a restatement of the coverage gap.

Table 7. Conformal performance by split regime and method, extremely randomised trees, nominal coverage 0.90. Marginal coverage is the mean over splits (ten for random, scaffold and cluster; twenty-one for family). The least-similar band is Tanimoto < 0.4, pooled over splits. Width is in K. Fallback is the mean fraction of test structures whose category had no calibration members and which therefore received the global quantile. CWC is the coverage-width criterion with $\eta = 30$. Source: results/table_conformal.csv and results/table_applicability_domain.csv.

Regime	Method	Coverage (mean of splits)	Band < 0.4 (pooled)	Width (K)	Fallback	CWC
Random	SCP	0.904	0.657	132.3	0.00	132
Random	Normalised SCP	0.901	0.759	127.0	0.00	127
Random	Mondrian, family	0.899	0.696	133.1	0.08	136
Random	Mondrian, similarity	0.907	0.888	129.1	0.00	129
Scaffold	SCP	0.834	0.552	126.5	0.00	923

Regime	Method	Coverage (mean of splits)	Band < 0.4 (pooled)	Width (K)	Fallback	CWC
Scaffold	Normalised SCP	0.825	0.629	124.1	0.00	1,165
Scaffold	Mondrian, family	0.832	0.576	127.2	0.10	990
Scaffold	Mondrian, similarity	0.895	0.938	154.8	0.00	179
Cluster	SCP	0.730	0.655	126.4	0.00	20,920
Cluster	Normalised SCP	0.711	0.667	127.7	0.00	37,007
Cluster	Mondrian, family	0.758	0.694	141.8	0.21	10,101
Cluster	Mondrian, similarity	0.886	0.825	232.2	0.13	356
Family	SCP	0.701	0.636	121.7	0.00	48,020
Family	Normalised SCP	0.709	0.650	129.5	0.00	39,813
Family	Mondrian, family	0.701	0.636	121.7	1.00	48,020
Family	Mondrian, similarity	0.877	0.890	208.3	0.00	411

Whether a global widening could have bought the same result is worth testing directly, because if it could the taxonomy would be doing no work. The oracle control of Section 2.7 multiplies every split conformal interval about its centre by a single constant k (Table 8, Figure 6). The constant can be chosen two ways, and the choice decides what the comparison means. Matching on mean *width* leaves the inflated predictor holding more marginal coverage than the novelty-conditioned one, 0.908 against 0.898 under family holdout and 0.905 against 0.870 under cluster, so a band comparison at that setting is not like for like: part of any advantage it shows is simply the extra coverage it was handed. Matching on marginal *coverage* fixes the coverage budget and asks only how each predictor allocates it across the similarity range, which is the question the taxonomy exists to answer. The coverage-matched comparison is the primary one here.

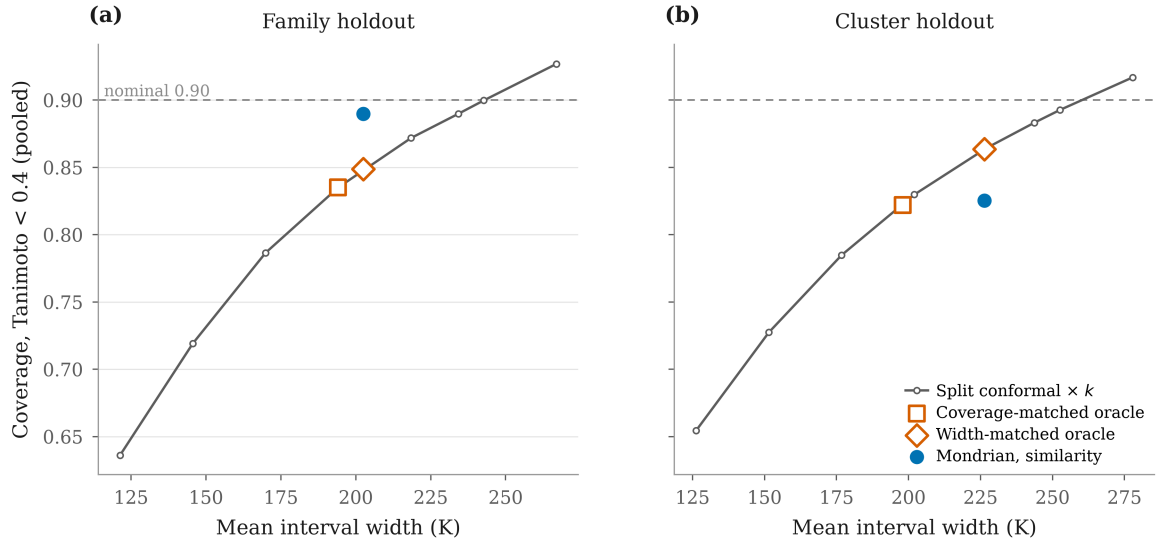


Figure 6. Coverage in the least-similar band (Tanimoto < 0.4), pooled over splits, against mean interval width in K , under (a) family and (b) cluster holdout. The grey line is split conformal with every interval multiplied about its centre by a constant k , swept from 1.0 to 2.2. The two open markers are oracle choices of k : the square reproduces the novelty-conditioned predictor's marginal coverage, the diamond equalises its mean width. Neither is knowable at prediction time. The filled point is the novelty-conditioned predictor itself. The square is the like-for-like comparison; reading the figure along the width axis alone compares predictors that do not hold the same marginal coverage, which is the confound Section 3.4 sets out. Source: results/table_width_matched.csv.

At matched marginal coverage the novelty-conditioned predictor is not beaten in either regime. Under family holdout the constant that reproduces its 0.898 pooled coverage is $k = 1.599$, and it covers 0.835 of the least-similar band (Tanimoto < 0.4 , 2,614 pooled predictions) against 0.890. Those are paired predictions on identical structures, so the discordant pairs settle the comparison: the novelty-conditioned predictor covers 164 structures the inflated one misses and misses 21 that it covers, McNemar $p = 1.1 \times 10^{-28}$. Under cluster holdout the matched constant is $k = 1.568$ and the two are indistinguishable in that band, 0.822 against 0.825, on 685 discordant pairs one way and 644 the other (McNemar $p = 0.27$). The taxonomy therefore wins decisively under family holdout, where it never falls back to the global quantile, and ties under cluster holdout, where 12.8% of test structures fall in a similarity band with no calibration members and receive the global quantile anyway (Table 7; Section 4.3).

What it wins is allocation and not efficiency, and that should be said plainly. At matched coverage the novelty-conditioned intervals are the *wider* of the two in both regimes: 202.5 K against 194.1 K under family holdout and 226.3 K against 197.9 K under cluster. A uniform inflation buys the same marginal coverage for less total width. What it cannot buy is the same distribution of that width over the similarity range.

A reader may reasonably ask the width-matched question as well, since width is the resource an interval spends, and the answer points the same way under family holdout. At equal mean width there, $k = 1.669$, the inflated predictor still covers only 0.849 of the least-similar band against 0.890 (134 discordant pairs against 27, McNemar $p = 3.0 \times 10^{-18}$), and it does so while over-covering marginally at 0.908. Matching the novelty-conditioned band coverage by inflation alone takes $k = 1.93$, which carries the mean interval from 203 K to 234 K; reaching nominal 0.90 in that band takes $k = 2.0$, at 243 K and 0.945 marginal. Under cluster holdout the equal-width constant $k = 1.792$ does reach 0.864 in that band against 0.825, but at 0.905 marginal coverage against 0.870, which is where the difference comes from: the coverage-matched row above removes it.

Both controls are oracles. The constant is computed from the test set it is then compared against, so none of 1.568, 1.599, 1.669, 1.792, 1.93 or 2.0 is knowable at prediction time, and it is regime-dependent, so a constant chosen on one holdout has no claim on another. Their role is to bound what any global inflation of split conformal could achieve, not to be deployed.

Table 8. Oracle inflation controls, extremely randomised trees, nominal coverage 0.90. Split conformal intervals are multiplied about their centre by the constant k . The coverage-matched row in each block is the k that reproduces the novelty-conditioned predictor's marginal coverage on the same test set and is the primary comparison; the width-matched row is the k that equalises mean width instead, and is handed more marginal coverage in the process. Neither is available at prediction time. Coverage is pooled over all held-out predictions in the regime, 7,174 under family holdout and 24,985 under cluster holdout; the least-similar band is Tanimoto < 0.4 and holds 2,614 and 12,846 of them. Width in K. The full sweep over k is in results/table_width_matched.csv.

Regime	Predictor	k	Coverage (pooled)	Band < 0.4 (pooled)	Width (K)
Family	Split conformal	1.000	0.743	0.636	121.4
Family	Coverage-matched oracle	1.599	0.898	0.835	194.1
Family	Mondrian, similarity	—	0.898	0.890	202.5
Family	Width-matched oracle	1.669	0.908	0.849	202.5
Family	Inflated to match band	1.930	0.939	0.890	234.2
Family	Inflated to nominal band	2.000	0.945	0.900	242.7
Cluster	Split conformal	1.000	0.716	0.655	126.3

Regime	Predictor	k	Coverage (pooled)	Band < 0.4 (pooled)	Width (K)
Cluster	Coverage-matched oracle	1.568	0.870	0.822	197.9
Cluster	Mondrian, similarity	—	0.870	0.825	226.3
Cluster	Width-matched oracle	1.792	0.905	0.864	226.3

3.5. An applicability domain, and an honest limit on the repair

Resolving coverage by similarity band under family holdout (Table 9) turns the result into an operational rule, and into an applicability-domain statement of the kind conformal prediction has been used to make since Norinder et al. [21]: the interval, rather than a separate distance threshold, carries the domain information. All coverage in this section is band-resolved: pooled within a band over the twenty-one splits, which together hold out each of the 7,174 structures exactly once. Split conformal covers 0.636 of structures whose nearest training analogue lies below Tanimoto 0.4, and 0.771, 0.809 and 0.874 in the next three bands. Its intervals are essentially constant in width (120.7–122.9 K) across all six bands, while the absolute error across those bands varies by a factor of 1.83, from 55.5 K in the least-similar band to 30.3 K in the 0.6–0.7 band, its lowest (Figure 7). Applying one width to a 1.8-fold spread in error is precisely the failure mode. Novelty-conditioned calibration instead holds 0.890, 0.919, 0.907 and 0.937 in the four least-familiar bands, at widths that scale from 246 K down to 149 K.

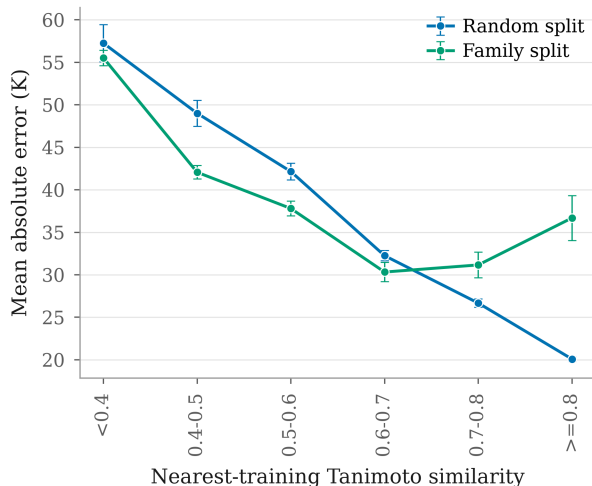


Figure 7. Mean absolute error in K against nearest-training similarity, for random and family-holdout splits, extremely randomised trees, pooled within each band. Error bars are the standard error of the mean of within-band absolute error. The random-split bars are anticonservative, because the ten repeats draw on the same 7,174 structures and a structure recurs across them, so the within-band observations are not independent. Band sizes are not annotated on the figure; they are in Table 9 for the family holdout and in results/table_band_mae.csv for both regimes.

The repair has a cost that should be stated plainly. In the most-similar band (Tanimoto ≥ 0.8) the novelty-conditioned predictor *under-covers*, at 0.673 against split conformal's 0.788, because it narrows those intervals to 88 K. Under a family holdout a test structure that closely resembles training data is unusual, and the calibration points populating that band come from families where high similarity genuinely did signal an easy prediction. The trade is nonetheless favourable in practice: that band holds 156 of 7,174 held-out predictions (2.2%), while the two least-similar bands hold 4,647 (64.8%). The recommendation is therefore to report novelty-conditioned intervals together with the nearest-training similarity itself, so that a reader can see which regime a given prediction sits in rather than trusting a single interval uniformly.

Table 9. Coverage, interval width and absolute error resolved by nearest-training Tanimoto similarity band under family holdout, extremely randomised trees, nominal coverage 0.90. The twenty-one splits hold out each of the 7,174 structures exactly once, so n sums to 7,174 and all

quantities are pooled within a band across splits. Widths and errors in K. Error is not monotone in similarity: it falls to 30.3 K in the 0.6–0.7 band and rises again above 0.8. Sources: results/table_applicability_domain.csv, results/table_band_mae.csv.

Band	<i>n</i>	MAE (K)	SCP coverage	SCP width (K)	Mondrian coverage	Mondrian width (K)
< 0.4	2,614	55.5	0.636	121.2	0.890	245.5
0.4–0.5	2,033	42.1	0.771	121.3	0.919	205.5
0.5–0.6	1,403	37.8	0.809	121.3	0.907	176.7
0.6–0.7	605	30.3	0.874	121.5	0.937	148.7
0.7–0.8	363	31.2	0.854	122.9	0.832	115.0
≥ 0.8	156	36.7	0.788	120.7	0.673	87.9

3.6. The penalty follows which structures were removed, not how many

The matched design separates the two effects that a conventional family holdout confounds (Figure 3, Table 10). Across twenty-one families and three repeats, with the test set held identical and the two comparison pools held identical in size:

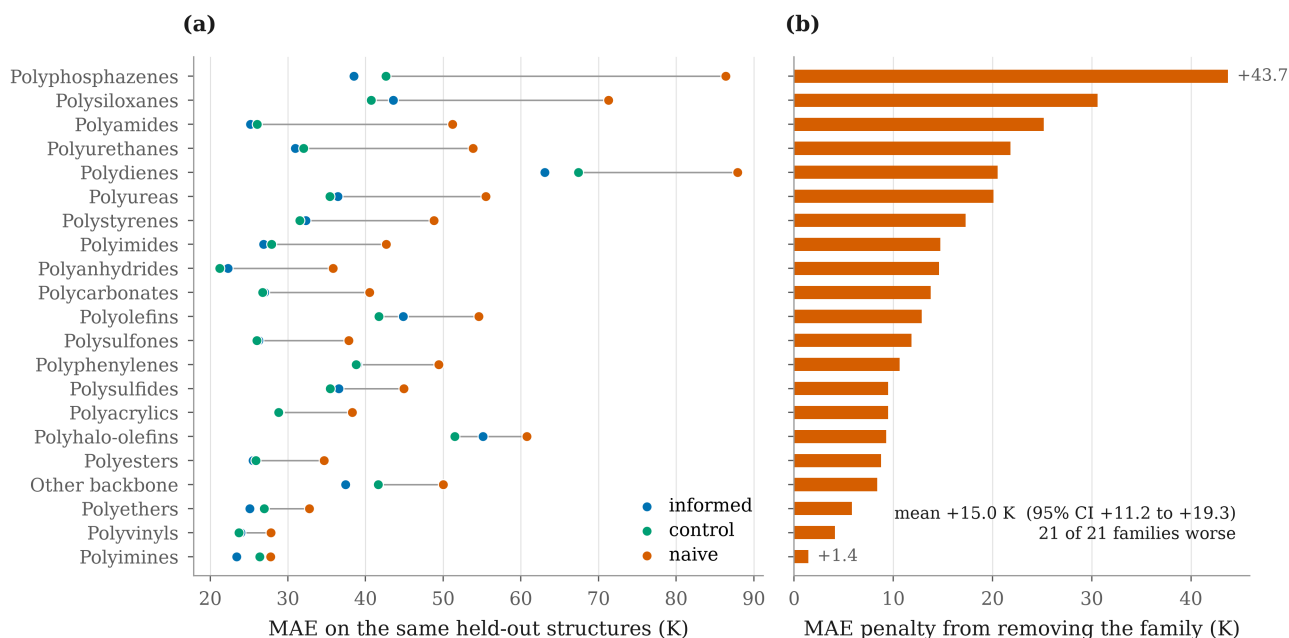


Figure 3. Matched-pair design, extremely randomised trees, twenty-one families with three repeats of the test draw and the control block in each. (a) The three training arms predicting an identical set of held-out structures, one row per family, each point the mean of that family's three repeats. Informed and control are coincident to within a few kelvin and of inconsistent sign (polydienes 63.1 against 67.4 K, polyhalo-olefins 55.1 against 51.5 K, polyolefins 44.9 against 41.7 K, largest separation 4.3 K), while naive is displaced to the right in every family. (b) The family effect, naive minus control, with training-set size held constant.

- Removing an equal quantity of *unrelated* structures — the **control** arm — changes MAE by +0.29 K (95% CI [−0.63, +1.24], Wilcoxon $p = 0.71$, worse in 9 of 21 families) for extremely randomised trees, and by −0.58 K (95% CI [−1.81, +0.64], $p = 0.45$, worse in 10 of 21) for histogram gradient boosting. The two models do not agree even on the sign.
- Removing *the family itself* — the **naive** arm — costs +14.97 K (95% CI [+11.21, +19.28]) for extremely randomised trees and +15.15 K (95% CI [+11.02, +20.03]) for histogram gradient boosting, and **is worse in all twenty-one families for both models**. With forty-two of forty-two signs positive the Wilcoxon test returns its floor, $p = 9.5 \times 10^{-7}$ per model; the sign count is the result, not the p -value.

Table 10. Matched-pair effects in K, over twenty-one families and three repeats per family, with the test set held identical within each triple. Family effect is naive – control, size effect is control – informed, total effect is naive – informed; positive means worse. Each family contributes the mean of its three repeats, the mean column averages those twenty-one family means, and the 95% interval is a percentile bootstrap over them, to be read as dispersion across families rather than inference about a population (Section 2.8). The Wilcoxon p of 9.5×10^{-7} is the floor of the test at twenty-one paired values, reached whenever all signs agree. Source: results/table_matched_statistics.csv.

Model	Effect	Mean (K)	95% CI (K)	Median (K)	Wilcoxon p	Positive
Extremely randomised trees	Family	+14.97	[+11.21, +19.28]	+12.86	9.5×10^{-7}	21 / 21
Extremely randomised trees	Size	+0.29	[−0.63, +1.24]	−0.15	0.71	9 / 21
Extremely randomised trees	Total	+15.26	[+11.35, +19.83]	+12.55	9.5×10^{-7}	21 / 21
Histogram gradient boosting	Family	+15.15	[+11.02, +20.03]	+11.28	9.5×10^{-7}	21 / 21
Histogram gradient boosting	Size	−0.58	[−1.81, +0.64]	−0.02	0.45	10 / 21
Histogram gradient boosting	Total	+14.57	[+10.63, +19.35]	+12.11	9.5×10^{-7}	21 / 21

The two effects are therefore not merely different in size but different in character. The family effect is roughly 15 K and unanimous in sign across every family and both models. The size effect is under 1 K in magnitude, has opposite signs in the two models, and scatters around zero per family, from -3.6 K to $+4.2$ K for extremely randomised trees and from -8.9 K to $+7.3$ K for histogram gradient boosting. We deliberately do not quote a ratio between them: with the denominator's sign unresolved, a ratio is not a quantity. What the design establishes is that the total effect, $+15.26$ K and $+14.57$ K for the two models, is accounted for by the family term alone, and it does so on matched test structures, so it is not an artefact of families differing in intrinsic difficulty or in T_g spread.

What the two arms actually differ in is the *locality* of the removal, not its volume: the naive arm deletes a connected region of structure space, the control arm a diffuse sample of the same size scattered across the other twenty families. The chemical reading of that contrast is developed in Section 4.2; the measurement is the locality contrast.

Family-level magnitudes rank as the chemistry predicts. For extremely randomised trees, polyphosphazenes suffer most ($+43.7$ K), followed by polysiloxanes ($+30.6$ K) and polyamides ($+25.2$ K); at the other end polyimines ($+1.4$ K), polyvinyls ($+4.1$ K) and polyethers ($+5.8$ K) are nearly unaffected, because the linkages that set their backbone mobility survive elsewhere in the training set. Note that polyamides rank third here despite being the second-largest family in the dataset (999 structures) — further evidence that abundance does not confer transferability when the chemistry is distinctive.

A near-duplicate objection has to be answered separately. The informed arm keeps the rest of the family, so for some test structures it holds a very close analogue that the naive arm cannot have, and the family effect might be no more than the loss of those analogues. Splitting the effect by how close the informed arm's nearest training neighbour actually was tests that directly (Table 11). The effect does not vanish where no near analogue existed. For extremely randomised trees it is $+11.4$ K over the 1,426 test structures whose nearest informed-arm neighbour lay below Tanimoto 0.6, against $+15.6$ K over the 1,655 whose neighbour lay above 0.9; for histogram gradient boosting the same two bands give $+11.6$ K and $+13.7$ K. The effect is

larger where a near-duplicate was available, which is what one would expect, but most of it is present where none was.

Table 11. Matched-pair family effect resolved by how close the informed arm's nearest training neighbour was, in K, over twenty-one families and three repeats. Bands are the informed arm's nearest-training Tanimoto similarity for the same test structure; n is the number of test structures in the band, each appearing once per repeat, paired across arms. Arm means are MAE in K over those structures, and the family effect is the paired mean of naive – control. Source: results/table_matched_by_analogue.csv.

Model	Informed-arm band	n	Informed	Control	Naive	Family effect
Extremely randomised trees	< 0.6	1,426	45.48	43.63	55.05	+11.42
Extremely randomised trees	0.6-0.8	2,307	29.89	30.44	46.18	+15.74
Extremely randomised trees	0.8-0.9	1,071	23.47	23.87	38.52	+14.65
Extremely randomised trees	≥ 0.9	1,655	17.83	20.48	36.09	+15.61
Histogram gradient boosting	< 0.6	1,426	43.95	41.63	53.27	+11.63
Histogram gradient boosting	0.6-0.8	2,307	29.35	29.28	42.61	+13.32
Histogram gradient boosting	0.8-0.9	1,071	23.88	24.03	36.10	+12.07
Histogram gradient boosting	≥ 0.9	1,655	18.46	20.12	33.83	+13.71

3.7. The gap is not an artefact of the descriptor set

A natural objection is that the generalisation gap reflects a limitation of RDKit descriptors rather than of the data, and that a representation better suited to novel chemistry would close it. Repeating the random and family-holdout regimes with binary Morgan fingerprints (radius 2, 2,048 bits) in place of the descriptor block tests this directly, since the two representations share almost nothing: one is a set of physically motivated aggregate quantities, the other a sparse record of local substructural environments. Both use extremely randomised trees. The two family arms are the full twenty-one holdouts, but the random arms differ in repeat count: five splits for Morgan against ten for descriptors.

The gap survives the substitution and is not narrower (Table 12). Fingerprints are worse in absolute terms in both regimes — 35.8 K against 28.5 K on random splits and 62.7 K against 49.6 K under family holdout — and their family-holdout penalty is larger in absolute terms (+26.9 K against +21.1 K) while being similar as a ratio (1.752 against 1.738). We do not attach an interval to the difference between those two ratios, because the random arms behind them rest on different numbers of repeats; the statement supported is that substructural environments offer no protection here in aggregate.

Table 12. Representation ablation, extremely randomised trees. MAE in K as the mean over splits, with the standard deviation across splits after $\pm$ and the number of splits in parentheses. Penalty is family MAE minus random MAE and the ratio is their quotient. Sources: results/table_representation_summary.csv, results/table_representation_ablation.csv.

Representation	Random MAE (K)	Family MAE (K)	Penalty (K)	Penalty ratio
RDKit descriptors, 217 columns	28.51 $\pm$ 0.93 (10)	49.56 $\pm$ 16.19 (21)	+21.05	1.738
Morgan, radius 2, 2,048 bits	35.81 $\pm$ 0.92 (5)	62.73 $\pm$ 15.38 (21)	+26.92	1.752

The per-family picture is not uniform, and one plausible mechanism is ruled out by it. Fingerprints are worse than descriptors on eighteen of the twenty-one withheld families, but they are *better* on three, and the largest single improvement is on polyphosphazenes (62.7 K against 87.1 K), the family whose absent P=N bits would be the obvious reason for a fingerprint model to fail. Nor are they better on polysiloxanes (77.6 K against 70.4 K). An explanation in terms of missing backbone bits therefore does not survive contact with the per-family data, and we do not offer one. What the ablation shows is narrower and still useful: the gap is not specific to the descriptor block, so it cannot be dismissed as an artefact of that feature set. Whether some other representation — a model pretrained on millions of hypothetical polymers, for instance — could narrow it is untested here and is the natural next step (Section 4.5).

3.8. The calibration set is a second lever, with a caveat

Sections 3.4 and 3.5 repair conditional coverage by changing how the conformal *quantile* is computed. A second possibility is to leave the quantile alone and change what the calibration set contains. Standard split conformal draws calibration points at random from the training pool, so every calibration residual describes an interpolation — the model had relatives of that repeat unit in training. Under family holdout the test residuals describe extrapolation, so the quantile is calibrated against the wrong population. Rebuilding the calibration set by holding families out *within* the training pool produces residuals of the kind the test set will actually demand.

Both levers work, and to a similar degree (Table 13).

Table 13. Calibration-set construction under family holdout, extremely randomised trees, nominal coverage 0.90, with the point model identical in all four arms. Coverage is the mean over the twenty-one splits. Worst band is the split-averaged worst band defined in Section 2.7: within each split the weakest similarity band holding at least ten structures is taken and those minima are then averaged, which is why it sits below the band-resolved coverage of the least-similar band in Table 9. Dropping the ten-structure restriction moves the first row from 0.564 to 0.539. Width in K, as a mean over splits; *n* calibration is the mean calibration-set size per split. Source: results/table_calibration_study.csv.

Calibration set	Quantile	Coverage (mean of splits)	Worst band (split-averaged)	Width (K)	<i>n</i> calibration
Random	Global (SCP)	0.701	0.564	122	1,366
Random	Novelty-conditioned	0.877	0.753	208	1,366
Family-out	Global (SCP)	0.855	0.753	187	3,349
Family-out	Novelty-conditioned	0.863	0.778	189	3,349

Rebuilding the calibration set lifts plain split conformal from 0.701 to 0.855 mean-of-splits without touching the quantile rule, the model or the representation. It does so by widening intervals from 122 K to 187 K, which is the appropriate response: the residuals it now calibrates against are genuinely larger. The construction is not, however, the only thing that changes in that swap. The family-out construction also enlarges the calibration set from a mean of 1,366 structures to 3,349, because residuals are pooled over the held-out training families rather than drawn once. A larger calibration set tightens the quantile's own sampling error, so the 0.701 $\rightarrow$ 0.855 improvement is attributable to the two changes jointly and this experiment cannot separate them.

The two repairs are largely substitutes rather than complements. Applying both gives 0.863 marginal coverage, no better than the novelty-conditioned quantile alone at 0.877, because they address the same root cause from opposite ends — that calibration residuals understate

extrapolation error. The combination has the best split-averaged worst band of the four arms, 0.778 against 0.753 and 0.753, but that margin is not resolved: across the twenty-one families the combination beats the novelty-conditioned quantile alone on worst-band coverage in only 9, with a mean difference of +0.025 and a Wilcoxon p of 0.40. It should be read as no worse rather than as better.

That comparison depends on how band coverage is aggregated, and the dependence should be disclosed. Under the band-resolved convention used in Table 9, the least-similar band gives 0.890 for the novelty-conditioned quantile alone, 0.824 for family-out split conformal and 0.876 for the two combined, against 0.636 for plain split conformal. Read that way the novelty-conditioned quantile is ahead of both family-out arms rather than level with them, and the statement that both levers work to a similar degree holds only under the split-averaged convention. The split-averaged figures are quoted here because they are the more conservative of the two against this section's own recommendation, which is to use the novelty-conditioned quantile.

The practical guidance follows the cost. Novelty-conditioned calibration is free: it reuses the same fitted model and calibration set, requiring only that similarity be computed. Family-out calibration requires refitting the model once per held-out training family — here eight additional fits per split. For most purposes the novelty-conditioned quantile alone is the better trade.

3.9. A Van Krevelen baseline deteriorates less and predicts worse

The additive schemes that machine-learned models propose to replace make a different inductive assumption: that T_g is a composition-weighted sum of group contributions, with no interaction terms and no dependence on which linkages the groups sit in. If the family-holdout penalty is really about absent linkage chemistry, a model that never learns linkage-specific behaviour in the first place should lose less when a linkage type is withdrawn. It does, and it pays for that in accuracy everywhere (Table 14).

Table 14. Van Krevelen group-contribution baseline against the ensemble, MAE in K as the mean over splits (ten for random, scaffold and cluster; twenty-one for family). Penalty is family MAE minus random MAE and the ratio is their quotient. The additive contributions Y_i are refitted on each training fold, so the baseline receives the same data as the ensemble. Source: results/table_baseline_comparison.csv.

Model	Random	Scaffold	Cluster	Family	Penalty (K)	Ratio
Group contribution	43.84	43.80	54.86	59.80	+15.96	1.364
Histogram gradient boosting	28.13	33.81	44.47	48.11	+19.98	1.710
Training-median baseline	93.79	95.71	121.61	95.46	+1.67	1.018

The group-contribution baseline of Section 2.6 gives 43.8 K MAE on random splits, 43.8 K under scaffold holdout, 54.9 K under cluster holdout and 59.8 K under family holdout. The absolute penalty, random to family, is +16.0 K against the ensemble's +20.0 K, and the ratio is 1.36 against 1.71. The flat random-to-scaffold step is the sharper observation: 43.84 K to 43.80 K, no change at all, where the ensemble loses 5.7 K. A model whose only inputs are group counts is indifferent to whether it has seen the scaffold before.

The honest way to read this is absolute first. The baseline is worse than the ensemble in every regime, by 15.7 K on random splits and by 11.7 K under family holdout, and its smaller penalty ratio partly reflects a worse starting point: it has less accuracy to lose. Resolved by family, it beats the ensemble on only two of the twenty-one withheld families, polysiloxanes (68.2 K against 69.7 K) and polysulfones (35.3 K against 36.7 K), both narrowly. It is not an alternative to the ensemble for prediction.

It is nonetheless independent support for the mechanism, from a model class with a different inductive bias and a different failure mode. Compositional additivity buys transferability at the cost of accuracy; the ensemble makes the opposite trade, and the size of the trade is what Sections 3.6 and 4.2 are about.

3.10. Sensitivity to the family taxonomy

Every family-dependent result inherits the labels, so the taxonomy is a legitimate target for a referee. During this work a boundary-split bug was found and fixed in the classifier: because the backbone was taken as the path between the two attachment points, a repeat unit written cut at its own characteristic linkage lost that linkage, so nylon-6 written `*NCCCCC(=O)*` read as a plain hydrocarbon chain and was labelled a polyvinyl. The dimer construction of Section 2.2 is the fix, and the same revision replaced a catch-all hydrocarbon assignment with the Other backbone abstention class. Together these reassigned 204 of the 7,174 structures, 2.8% of the dataset, the largest movements being 88 polyesters that are really polycarbonates, 46 polyolefins carrying a backbone heteroatom, 23 polyethers that are polyesters, and 15 polyvinyls that are polyamides.

Rerunning the whole pipeline against both label sets gives a direct sensitivity measurement. The random, scaffold and cluster results are unchanged to three decimal places for every model, as they must be, since those regimes never consult a family label; pooled R^2 in those three regimes is likewise identical to three decimals. Under family holdout, where the labels do enter, pooled R^2 moves by at most 0.006 across the five models — the largest is support vector regression at 0.0052 — and is unchanged at 0.746 for histogram gradient boosting. The remaining family-dependent results move by about a kelvin and no conclusion changes sign: family-holdout MAE goes from 46.8 K to 48.1 K for histogram gradient boosting and from 48.9 K to 49.6 K for extremely randomised trees, and the matched-design family effect from +14.61 K to +14.97 K and from +13.28 K to +15.15 K. The 1.29 K move in family-holdout MAE has two parts, because the revision also created a twenty-first holdout group. Averaged over the same twenty groups the old labels supplied, the corrected labels give 47.89 K, so relabelling accounts for +1.07 K of the move and the new Other backbone group for the remaining +0.22 K. The one qualitative change is in the paper's favour. Under the old labels the family effect was positive in 20 of 20 families for one model but only 18 of 20 for the other, the two exceptions being polycarbonates and polysulfides; polycarbonates was also the family most altered by the correction, nearly doubling from 98 to 189 structures as the misassigned polyesters returned to it. Under the corrected labels the effect is positive in 21 of 21 families for both models. The earlier anomaly was a labelling error, not a counterexample.

This is a sensitivity check against one specific perturbation, not a proof of robustness to any taxonomy. A chemically different but defensible scheme — one that split polyimides by dianhydride class, say — would move the family-level numbers again. What the check establishes is the order of magnitude of that movement for a 2.8% relabelling: about a kelvin on aggregate errors, with the sign structure of the matched design intact.

4. Discussion

The result a polymer chemist can act on is the family-resolved one. A T_g model does not have a single accuracy: histogram gradient boosting predicts withheld polyvinyls to 30.2 K and withheld polyphosphazenes to 84.8 K, and it displaces whole backbone classes in a consistent direction rather than scattering about them. What follows takes that structure apart: what an aggregate leaderboard number conceals (Section 4.1), why the penalty follows which chemistry is missing rather than how much data is missing (Section 4.2), why an interval conditioned on structural novelty survives where one conditioned on polymer family cannot (Section 4.3), what to do about it (Section 4.4), and what the evidence here does not support (Section 4.5).

4.1. What a T_g leaderboard is actually measuring

The most consequential number in this study is not an error but a contrast. Changing the model changes MAE by 0.38 K between the two best and by 2.74 K across all four non-trivial regressors; changing the partitioning protocol from random to family holdout changes it by 20.0 K. Any comparison of architectures on this dataset that does not fix and report the split regime is dominated by a variable it does not control.

This reframes a decade of incremental gains. Reported improvements from kernel methods to tree ensembles to graph networks to pretrained chemical language models are real, but they are improvements in interpolation, measured where structural analogues are plentiful. None of the comparisons establishes that the newer model extrapolates better, because none was evaluated under a regime that requires extrapolation. That is not a criticism of the models; it is a criticism of the protocol they were scored under, and it is cheap to fix.

4.2. Why chemistry and not volume

The matched design measures a locality effect: at fixed training-set size, deleting a coherent chemical neighbourhood costs about 15 K on that neighbourhood, deleting a diffuse sample of the same size costs nothing that survives a change of model. The chemical reading of that asymmetry is the straightforward one. Descriptor-based models of T_g learn how particular backbone linkages translate into chain stiffness and interchain interaction, and those mappings are local to a linkage type. Remove every phosphazene, and no amount of additional polyester data teaches the model what a P=N backbone does to segmental mobility; it extrapolates organic intuition into inorganic chemistry and overpredicts those chains by 84 K on average (Section 3.3). Four lines of evidence support that reading, one of them only in part. Deterioration correlates with structural isolation while no size effect is resolved (Section 3.3). The family effect is not the loss of near-duplicates: it is +11.4 K even among test structures for which the informed arm itself held no training neighbour above Tanimoto 0.6 (Section 3.6). An additive model that never learns linkage-specific behaviour deteriorates proportionally less when a linkage type is withdrawn (Section 3.9). The fourth line, that the withheld-family errors are systematic displacements of a consistent sign, holds only partly, because for the families with the most extreme T_g regression toward the training mean predicts the same sign. It rests on the families that move against that line: polydienes, polyolefins, polyhalo-olefins and polystyrenes, all underpredicted while sitting below the dataset median (Section 3.3).

The consequence for data-collection strategy needs stating carefully, because the experiment removes data rather than adding it. What was tested is that withdrawing a median of 1.85% of the training pool at random, from chemistry already covered, does not measurably change accuracy on a held-out family, while withdrawing the same quantity as a coherent family does. That says a marginal fraction of the existing pool carries little information about a new backbone. It does not establish that *adding* data within covered chemistry would be useless, which is a different experiment at a different scale and one these results cannot settle. The defensible version of the recommendation is the comparative one: at equal cost, structures from an uncovered family are worth more than structures from a covered one, and a screening programme that plans its data collection around coverage of the structure space rather than around volume is acting on what was measured.

4.3. Why novelty-conditioning works and family-conditioning cannot

The contrast between the two Mondrian taxonomies is a structural fact rather than an empirical accident, and it generalises beyond this dataset. A conditional conformal predictor can only offer category-specific validity for categories it has calibration data for. Polymer family fails this test in exactly the deployment scenario of interest: the family is new, so it has no calibration members, and the predictor silently degrades to the unconditional one, here for 100% of test structures. Nearest-training similarity is computable for any repeat unit against any training set, its value is available before the label is, and its bands are populated by calibration structures drawn from every family, so the taxonomy remains well defined precisely where the family-based

one collapses. The same argument recommends similarity — or any other deployment-computable novelty coordinate — over class-based taxonomies for conditional conformal prediction in materials problems generally, wherever the classes of interest are the ones the model has not seen.

The similarity taxonomy is not, however, free of the same defect in kind. A similarity band can also be empty of calibration members, and under cluster holdout it is: 12.8% of test structures fall back to the global quantile there, against 21.5% for the family taxonomy. The difference is one of degree and of prognosis. A family holdout empties the relevant family category by construction, every time, and no amount of additional calibration data fixes it; a similarity band is emptied only when the calibration pool happens to contain nothing that novel, which calibration data drawn more widely can in principle remedy. The advantage is real but it should be stated as a smaller and repairable blind spot rather than as none.

Section 3.4 tested whether a global widening explains the advantage, and at matched marginal coverage it does not. An oracle inflation of split conformal set to the same marginal coverage covers 0.835 of the least-similar band under family holdout against 0.890, losing the paired comparison decisively (McNemar $p = 1.1 \times 10^{-28}$), and 0.822 against 0.825 under cluster holdout, which is a tie ($p = 0.27$). The taxonomy is ahead where it never falls back and level where it falls back for 12.8% of structures. What it buys is allocation rather than efficiency: at matched coverage its intervals are the wider of the two, 202.5 K against 194.1 K under family holdout, so the claim is about where the width goes and not about spending less of it.

A weighted conformal correction is the other principled response to a covariate shift of this kind [30], and it is not used here for a practical reason. The weights it needs are the likelihood ratio between the test and training covariate distributions, estimated from unlabelled test data. With 44 to 1,707 structures per family in a 217-dimensional descriptor space, and with the shift being the wholesale absence of a chemistry rather than a reweighting of a shared support, that ratio is not estimable at the sample sizes available. A novelty-conditioned taxonomy asks less of the data: it needs only a coordinate that is computable at prediction time and calibration members in each band.

4.4. Practical recommendations

For anyone building or reporting a T_g model intended for screening:

- 1. Report the split regime as a primary result, not a methods detail,** and report at least one chemistry-aware regime alongside any random split. The gap between them is the honest estimate of screening performance.
- 2. Repeat every regime that can be repeated, and say what the spread means.** Scaffold and cluster splits vary by ± 8 –10 K across repeats. A family holdout is run once per family, so its ± 15 K is dispersion between families, not run-to-run noise; both are worth reporting, and conflating them is not.
- 3. Report errors resolved by polymer family, with the sign of the error.** An aggregate MAE of 48 K conceals a range from 30 K to 85 K, and a user working on polysiloxanes is served by the wrong number; the -64 K bias on that family tells them more than either.
- 4. **Ship intervals, and condition them on a deployment-computable novelty coordinate.**** Report the nearest-training similarity next to each prediction so a reader can see which regime it belongs to.
- 5. Fit every data-dependent preprocessing step inside the fold,** including descriptor filtering. It is a one-line change to a pipeline and removes an unquantified optimism from the reported numbers.

4.5. Limitations

The target itself is noisy in a way no structural model can address. Independent literature reports of the same repeat unit differ by up to 115 K in this dataset, because a repeat-unit graph cannot express molecular weight, tacticity, crosslink density, thermal history or measurement protocol. A 28 K random-split MAE should be read against that floor, and part of the residual is irreducible.

The family taxonomy is our own. Section 3.10 measures its sensitivity to one 2.8% relabelling and finds about a kelvin of movement with every conclusion intact, but that is one perturbation, not a survey of chemically defensible alternatives. The validation in Table S1 should also be described accurately: it is a pinned regression suite, and its twenty-five cases are the ones the classifier's patterns were written to handle. It protects against future regressions and it demonstrates that the rules encode the chemistry we claim they encode; it is not an independent held-out validation set, and it cannot be read as an accuracy estimate for the taxonomy on unseen chemistry.

The matched design removes data rather than adding it, and removes a median of 1.85% of the pool in the control arm (Section 2.5). Its conclusions are about where a fixed data budget should be spent, not about the returns to enlarging it.

Our models are descriptor-based ensembles and one additive baseline. We show in Section 3.7 that the generalisation gap survives a change of representation to Morgan fingerprints, but we have not tested a pretrained chemical language model or a graph neural network. Such a model, having seen millions of hypothetical polymers during pretraining, might have effective coverage of families absent from this dataset's *labelled* portion. Testing whether pretraining narrows the family effect is the natural next experiment, and the matched design transfers to it unchanged.

The signed-error evidence for the chemical mechanism is partial. For families whose T_g sits far from the rest of the dataset, regression toward the training mean and an absent linkage chemistry predict displacement of the same sign, and the twenty-one family means available here do not separate them (Section 3.3). The argument rests on the families that move against the shrinkage line; a design that varied T_g offset and backbone novelty independently would settle it, and this one does not.

Finally, the novelty-conditioned predictor under-covers the most-similar band under family holdout (Section 3.5) and falls back to the global quantile for 12.8% of structures under cluster holdout, where it holds no advantage over an oracle inflation of split conformal at matched marginal coverage (Section 3.4). It is also the wider of the two at matched coverage in both regimes, so what it offers is a better allocation of width rather than less of it. The affected groups are small and the direction of the trade favours screening, but the method is an improvement rather than a solution, and conditional coverage should continue to be reported rather than assumed.

5. Conclusions

Machine-learned glass transition temperatures are far more reliable inside familiar backbone chemistry than outside it, and standard practice measures only the former. On 7,174 experimental repeat units, moving from random to chemistry-aware partitions raises the error of the best model from 28.1 K to as much as 48.1 K and lowers pooled R^2 from 0.87 to 0.75. That 20.0 K shift is roughly fifty times the difference between the best and second-best model and seven times the spread across all four non-trivial regressors.

A matched-pair design shows that the penalty follows the locality of what is removed rather than its quantity: deleting an equal number of unrelated structures moves MAE by under 1 K, with the two models disagreeing on the sign, while deleting the family costs about +15 K and does so in all twenty-one families for both models. Transferability tracks how structurally isolated a family is; family size is not resolved at this sample size, and the two are nearly orthogonal, so the first result is not the second in disguise. An additive group-contribution baseline, which never learns

linkage-specific behaviour, loses proportionally less across the same regimes and predicts worse everywhere, which is the same mechanism seen from a model class with a different inductive bias.

The uncertainty attached to these predictions fails in the same place. Split conformal intervals hold their nominal 90% marginally on random splits while covering 66% of the least familiar structures, and lose marginal validity under shift. Conditioning the calibration on polymer family cannot repair this, because a withheld family has no calibration data by construction. Conditioning on nearest-training structural similarity does, holding 0.88–0.91 as a mean over splits in every regime tested and allocating width to the bands where the error actually is: at matched marginal coverage it covers more of the least familiar band than a uniform inflation of split conformal under family holdout, 0.890 against 0.835, and as much under cluster holdout, though it spends more total width to do so. Rebuilding the calibration set from within-training family holdouts is a second lever that reaches 0.855 with the plain global quantile, though in this experiment it also enlarges the calibration set from 1,366 to 3,349 structures, so construction and size cannot be separated; combining the two levers is no better than the novelty-conditioned quantile alone.

The practical implication is a short list: report a chemistry-aware split, repeat what can be repeated, resolve errors by family and by sign, and attach novelty-conditioned intervals together with the similarity value itself. None of these is expensive, and together they turn a T_g model from something that scores well into something a screening campaign can act on.

Supplementary Materials

The following are available in the repository named in the Data Availability Statement: Table S1, the twenty-five reference polymers with expected and assigned family; Table S2, family-holdout deterioration against structural isolation, T_g shift, T_g interquartile ratio and family size, with Pearson and Spearman correlations; Table S3, attachment-point counts per structure; Table S4, per-band coverage under each calibration construction; Table S5, absolute error resolved by nearest-training similarity band with the worst-to-best band ratio; Table S6, per-family group-contribution error under family holdout.

Author Contributions

Conceptualisation, R.S.; methodology, [FIRST AUTHOR INITIALS] and B.L.; software, R.S. and B.L.; validation, R.S. and L.E.; formal analysis, R.S.; investigation, R.S.; data curation, R.S. and L.E.; writing—original draft preparation, R.S. and B.L.; writing—review and editing, R.S., B.L. and L.E.; visualisation, B.L.; supervision, R.S.; project administration, R.S.. All authors have read and agreed to the published version of the manuscript.

Funding

This research received no external funding.

Institutional Review Board Statement

Not applicable.

Informed Consent Statement

Not applicable.

Data Availability Statement

All code, the curated structure-level table and every result file underlying this work are available at <https://github.com/Rugvedhs/spectrasense>, in the directory `polymer-tg-benchmark`. The

underlying experimental T_g records are redistributed with the POINT² benchmark [11] and derive from PoLyInfo [4].

Acknowledgments

The authors thank the maintainers of the POINT² benchmark and of PoLyInfo for making the underlying experimental records available, and the RDKit and scikit-learn developers for the software this work is built on.

Conflicts of Interest

The authors declare no conflict of interest.

Abbreviations

The following abbreviations are used in this manuscript:

T_g	glass transition temperature
MAE	mean absolute error
RMSE	root mean squared error
CI	confidence interval
SCP	split conformal prediction
CWC	coverage-width criterion
SMILES	simplified molecular-input line-entry system
PSMILES	polymer SMILES, with attachment points marked
SMARTS	SMILES arbitrary target specification
DSC	differential scanning calorimetry
DMA	dynamic mechanical analysis

Appendix A. Full per-family listing

Table A1. Family-holdout error and deterioration for histogram gradient boosting, all twenty-one families, in full. MAE and RMSE in K; deterioration is the family's holdout MAE over the model's mean random-split MAE. Table 5 quotes the three most and three least deteriorated rows of this listing. Source: results/table_family_holdout.csv.

group	n	mae	rmse	r2	deterioration	mean nn similarity
Polyphosphazenes	118	84.839	95.664	-1.339	3.016	0.304
Polydienes	77	79.648	107.076	0.020	2.831	0.367
Polysiloxanes	209	69.701	81.998	-0.368	2.478	0.388
Polyureas	44	61.770	73.913	0.312	2.196	0.506
Polyolefins	126	61.626	82.394	0.418	2.191	0.397
Other backbone	58	52.457	65.384	-0.783	1.865	0.403
Polyhalo-olefins	56	51.667	71.721	0.141	1.837	0.333
Polyurethanes	189	47.890	62.052	-0.192	1.702	0.477
Polyamides	999	46.924	62.150	0.556	1.668	0.513
Polyphenylenes	423	46.370	63.903	0.712	1.648	0.463
Polyanhydrides	53	43.798	54.665	0.302	1.557	0.489
Polystyrenes	250	42.206	54.871	0.312	1.500	0.399
Polysulfides	60	42.020	56.747	0.746	1.494	0.516
Polyimides	1707	37.759	49.749	0.584	1.342	0.406
Polysulfones	232	36.654	49.865	0.565	1.303	0.561
Polyimines	52	36.418	55.746	0.225	1.295	0.532
Polyacrylics	661	36.249	49.700	0.382	1.289	0.412
Polyesters	761	34.929	46.497	0.728	1.242	0.521
Polycarbonates	189	33.745	45.569	0.741	1.200	0.558
Polyethers	713	33.362	46.211	0.816	1.186	0.553
Polyvinyls	197	30.217	40.312	0.497	1.074	0.430

References

Numbered in order of first appearance, as Polymers requires.

1. Van Krevelen, D. W.; te Nijenhuis, K. *Properties of Polymers: Their Correlation with Chemical Structure; Their Numerical Estimation and Prediction from Additive Group Contributions*, 4th ed.; Elsevier: Amsterdam, **2009**.
2. Bicerano, J. *Prediction of Polymer Properties*, 3rd ed.; Marcel Dekker: New York, **2002**.
3. Askadskii, A. A. *Computational Materials Science of Polymers*; Cambridge International Science Publishing: Cambridge, **2003**.
4. Otsuka, S.; Kuwajima, I.; Hosoya, J.; Xu, Y.; Yamazaki, M. PoLyInfo: Polymer Database for Polymeric Materials Design. In *2011 International Conference on Emerging Intelligent Data and Web Technologies*; IEEE, **2011**; pp 22-29.
5. Tao, L.; Varshney, V.; Li, Y. Benchmarking Machine Learning Models for Polymer Informatics: An Example of Glass Transition Temperature. *J. Chem. Inf. Model.* **2021**, *61* (11), 5395-5413. DOI: 10.1021/acs.jcim.1c01031
6. Casanola-Martin, G. M.; Karuth, A.; Pham-The, H.; et al. Machine learning analysis of a large set of homopolymers to predict glass transition temperatures. *Commun. Chem.* **2024**, *7*, 226. DOI: 10.1038/s42004-024-01305-0
7. Qiu, H.; et al. Design of Polyimides with Targeted Glass Transition Temperature Using a Graph Neural Network. *J. Mater. Chem. C* **2023**. DOI: 10.1039/D2TC05174E
8. Kuenneth, C.; Ramprasad, R. polyBERT: a chemical language model to enable fully machine-driven ultrafast polymer informatics. *Nat. Commun.* **2023**, *14*, 4099. DOI: 10.1038/s41467-023-39868-6
9. Xu, C.; Wang, Y.; Barati Farimani, A. TransPolymer: a Transformer-based language model for polymer property predictions. *npj Comput. Mater.* **2023**, *9*, 64. DOI: 10.1038/s41524-023-01016-5
10. Kunchapu, S.; Jablonka, K. M. PolyMetriX: an ecosystem for digital polymer chemistry. *npj Comput. Mater.* **2025**, *11*, 312.
11. Xu, J.; et al. POINT²: A Polymer Informatics Training and Testing Database. *arXiv* **2025**, arXiv:2503.23491.
12. Sheridan, R. P. Time-Split Cross-Validation as a Method for Estimating the Effectiveness of QSAR Models. *J. Chem. Inf. Model.* **2013**, *53* (4), 783-790.
13. Wu, Z.; et al. MoleculeNet: a benchmark for molecular machine learning. *Chem. Sci.* **2018**, *9*, 513-530.
14. Babbar, A.; et al. Explainability and Transferability of Machine Learning Models for Predicting the Glass Transition Temperature of Polymers. *arXiv* **2023**, arXiv:2308.09898.
15. Akdoğan, M. Trust Beyond Accuracy: Conformal Uncertainty Quantification Reveals the Generalization Gap in Polymer Glass Transition Temperature Prediction. *ACS Omega* **2026**, *11* (30), 44926-44941.
16. Teh, S. X. Polymer Glass-Transition Temperature Prediction and Polymer-Family Generalisation. Software and report, GitHub repository. <https://github.com/tehsongxuan/Polymer-Tg-Prediction-and-Polymer-Family-Generalisation> (accessed 14 September 2026).
17. Papadopoulos, H.; Proedrou, K.; Vovk, V.; Gammerman, A. Inductive Confidence Machines for Regression. In *Machine Learning: ECML 2002*; LNCS 2430; Springer, **2002**; pp 345-356.
18. Vovk, V.; Gammerman, A.; Shafer, G. *Algorithmic Learning in a Random World*; Springer: New York, **2005**.
19. Lei, J.; G'Sell, M.; Rinaldo, A.; Tibshirani, R. J.; Wasserman, L. Distribution-Free Predictive Inference for Regression. *J. Am. Stat. Assoc.* **2018**, *113* (523), 1094-1111.

20. Agrawal, V. Polymer property prediction using ensemble of directed message-passing neural networks with uncertainty estimation and applicability domain analysis. *MRS Commun.* **2026**. DOI: 10.1557/s43579-026-00948-5
21. Norinder, U.; Carlsson, L.; Boyer, S.; Eklund, M. Introducing Conformal Prediction in Predictive Modeling. A Transparent and Flexible Alternative to Applicability Domain Determination. *J. Chem. Inf. Model.* **2014**, *54* (6), 1596–1603. DOI: 10.1021/ci5001168
22. RDKit: Open-source cheminformatics. <https://www.rdkit.org>
23. Rogers, D.; Hahn, M. Extended-Connectivity Fingerprints. *J. Chem. Inf. Model.* **2010**, *50* (5), 742–754.
24. Pedregosa, F.; et al. Scikit-learn: Machine Learning in Python. *J. Mach. Learn. Res.* **2011**, *12*, 2825–2830.
25. Bemis, G. W.; Murcko, M. A. The Properties of Known Drugs. 1. Molecular Frameworks. *J. Med. Chem.* **1996**, *39* (15), 2887–2893.
26. Barber, R. F.; Candès, E. J.; Ramdas, A.; Tibshirani, R. J. Predictive inference with the jackknife+. *Ann. Stat.* **2021**, *49* (1), 486–507.
27. Papadopoulos, H.; Haralambous, H. Reliable prediction intervals with regression neural networks. *Neural Netw.* **2011**, *24* (8), 842–851.
28. Vovk, V. Conditional validity of inductive conformal predictors. *Mach. Learn.* **2013**, *92*, 349–376.
29. Boström, H.; Johansson, U. Mondrian conformal regressors. *Proc. Mach. Learn. Res. (COPA)* **2020**, *128*, 114–133.
30. Tibshirani, R. J.; Barber, R. F.; Candès, E. J.; Ramdas, A. Conformal Prediction Under Covariate Shift. In *Advances in Neural Information Processing Systems 32 (NeurIPS 2019)*; **2019**. arXiv:1904.06019